

Executable Epistemic Contracts in Deterministic Agent Simulation: The ENIGMA Architecture¹

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September 17, 2026

Abstract

In most modern large language model (LLM) based agent simulations, agent beliefs are implemented as prompts or verbalization structures, and belief divergence arises as a side effect of stochastic text generation rather than as a designed, verifiable property. We present ENIGMA, a deterministic multi-agent architecture in which belief divergence is a first-class, causally efficacious, and executably verifiable property of the simulation. The core of the architecture consists of: ExperienceTrace, a unit of interpreted experience with first-class provenance (perception, testimony, introspection); a perception membrane that strictly separates world truth from agent observations; DeltaGate, the single legal “interpretation $\rightarrow$ state” entry point that isolates the LLM from the causal core; and BeliefCrystallizationEngine with asymmetric trauma ($\times 6$) and entropy decay ($\tau = 100$ ticks). The methodological contribution is executable epistemic contracts (SUPERBOX): scenarios that turn epistemic claims about the system into binary, reproducible verdicts. We report reproducible runs under a fresh-extraction protocol: 45/45 core invariant probes, deterministic belief revision, the measured update asymmetry, the propagation of a single rumor with listener-dependent confidence, and the belief-to-decision causality confirmed by a positive control—four agents witnessing a theft form beliefs with confidence 0.9, disposition-specific decision modifiers, and a changed intent. We also present an honest audit of the scenarios that fail in the studied snapshot. The red verdicts are not concealed; they demonstrate that the contracts function as a diagnostic instrument rather than decoration. The work is positioned strictly as a systems contribution and a runnable reference implementation, not as a new psychological theory.

Keywords: deterministic agent simulation; epistemics; belief revision; executable contracts; provenance; LLM agents; Theory of Mind.

1 Introduction

LLM-based agent simulations are growing rapidly: from the “generative agents” of Park et al. [1] with twenty-five characters in a sandbox environment to social simulators at the scale of a million agents [2]. These systems ascribe memory, reflection, plans, and beliefs to agents, and the resulting behavior often looks convincing. At the architectural level, however, a belief in such systems is usually text in a prompt or a record in a language memory: it influences the next LLM utterance, but it is not a state whose dynamics obey verifiable rules. Divergence of beliefs between agents—the central mechanism of social drama, lies, rumors, and misunderstanding—arises there as a side effect of generation stochasticity, not as a designed and controlled property of the system.

The consequence is epistemic unverifiability. The claim “agent A believes that B stole the ring” in an LLM-centric architecture can only be confirmed by re-querying the model, whose answer is unstable and

¹The complete source code is available at <https://github.com/4Um4/Enigma/> (version V.0.5.4.0.6). All numerical results in Section 5 were obtained by running this version in an isolated environment without an LLM server; reproduction commands are listed in the Availability section.

prompt-dependent. One cannot reproduce the moment a belief was formed, cannot recover its provenance (did the agent witness the event, or learn of it from a third party), and cannot prove that the belief, rather than generation noise, changed the agent’s decision. For research in which belief divergence is the object of study rather than decoration, this is insufficient.

This paper describes ENIGMA, a research prototype of a deterministic causal simulation engine in which epistemic divergence between agents is a first-class property. The main thesis is this: *we present a deterministic multi-agent architecture in which belief divergence is a first-class, causally efficacious property of the simulation, and in which every epistemic claim about the system is turned into a binary, reproducible verdict by an executable scenario. The scenarios survive persistence and session restart.* Determinism here does not make the world predictable for the player—it makes the epistemics inspectable for the researcher: the player sees the world through their own epistemic boundary, and the hidden state of other agents makes their behavior informationally unpredictable while remaining mechanically reproducible.

The contributions of this work are:

1. **Ontology and provenance.** A three-part ontology “world truth — observation — belief” in which the unit of experience, ExperienceTrace, carries first-class provenance (PERCEPTION, TESTIMONY, INTROSPECTION), and testimony admits *owner* $\neq$ *actor* (Section 3).
2. **Perception membrane and DeltaGate.** Architectural isolation of the LLM from the causal core: model language cannot become a source of truth; every interpretation passes a whitelist, clamps, and idempotency control (Section 3).
3. **An asymmetric belief crystallization model.** A measurable update dynamics with a trauma multiplier of 6, exponential decay, and a forgetting threshold, formalizing negativity bias in computable form (Section 3).
4. **Executable epistemic contracts.** The SUPERBOX scenario suite as an evaluation method: every architectural claim receives a binary, reproducible verdict; a red verdict is treated as diagnosis rather than concealment (Sections 4, 5).
5. **An honest audit.** Confirmed verdicts of both signs: sixteen green checks (including the positive control of belief-to-decision causality through the observation channel, and an accusation gate in which the player’s belief is a gameplay capability) and red verdicts with a precise diagnosis of every chain break (Section 5).

We deliberately position the work as a systems contribution and a runnable reference implementation, not as a new algorithm or psychological theory. Claims in the text are bound to specific code modules and specific scenarios; “a class name is not proof of semantics” is a working principle of the project, and we adhere to it throughout.

2 Related Work

LLM-based agent simulators. Generative agents [1] combine a memory stream, reflection, and planning; agent state is textual, and a “belief” exists only insofar as it is reflected in the memory from which the prompt is assembled. OASIS [2] scales this approach to a million agents in a structured social environment, but agent epistemics remains a projection of text. The system closest to our problem is BeliefNest [3]: an open-source simulator for embodied agents in Minecraft with explicit nested belief states, including false-belief tasks. There, however, belief representations serve prompt generation and the evaluation of LLM Theory-of-Mind reasoning, whereas in ENIGMA a belief is a state of the deterministic core whose formation, provenance, and causal role are controlled by engine contracts. Surveys of LLM-based agents [4] confirm that an epistemic layer with guarantees is rare: it is either absent or implemented inside the model itself.

Belief revision and the psychology of asymmetry. The classical AGM theory of belief revision [5] postulates rationality postulates for changing beliefs upon new information; our revision operation is

simpler (reliability weighting of the source, echo damping of repetitions) but, unlike AGM, it executes as part of the simulation tick and is therefore verifiable on live trajectories. The psychological literature documents a stable negativity asymmetry: negative information is stronger than positive information of equal intensity [6, 7]. The trauma multiplier of 6 in BeliefCrystallizationEngine is a deliberate engineering formalization of this effect: refuting trust must cost more than confirming it.

Theory of Mind and false beliefs. The classics of ToM [8] and the false-belief paradigm [9] provide a ready-made form of check: an agent who did not witness an event must demonstrate a belief that diverges from the truth. Our reference scenario “Double Truth” (Section 4) is a computational analogue of the Sally–Anne task: three agents, one episode, three different epistemic worlds, and the divergence must translate into divergent decisions.

Contract engineering. The idea of design by contract [10] and runtime verification [11] are transferred here to epistemic properties: postconditions and invariants correspond to executable SUPERBOX scenarios, and contract violations correspond to runtime exceptions (`ArchitecturalViolationError`) and red scenario verdicts. Unlike traditional runtime verification, where a monitor observes the system from outside, our contracts are built into the development loop as acceptance gates: a subsystem counts as implemented only when its scenario is green—the principle “claim $\rightarrow$ code $\rightarrow$ call path $\rightarrow$ test $\rightarrow$ observable result”.

Cognitive architectures. Systems such as Soar [12] have for decades implemented procedural memory and rules as executable structures. ENIGMA does not claim Soar-level generality; it studies a narrow but poorly illuminated question: how to make the *epistemic state* of an agent (what it knows, from where, and how confidently) an equally engineer-verifiable subsystem as working memory or a planner.

3 The ENIGMA Architecture

3.1 Ontology: truth, observation, belief

The engine never collapses three categories into one state: *world truth* (what happened in the simulation), *observation* (what an agent physically had access to), and *belief* (what an agent currently believes). Formally, the simulation state at tick t is

$$\mathcal{S}_t = (W_t, \{S_i^t\}_{i=1}^N, \mathcal{E}_t, \mathcal{C}_t),$$

where W_t is the world (state snapshot and objects), S_i^t is the state of agent i (body, memory, perception, relationships, psyche), $\mathcal{E}_t$ is the epistemic records of beliefs, and $\mathcal{C}_t$ is active commitments. The transition between ticks is given by the operator

$$\mathcal{S}_{t+1} = \Phi(\mathcal{S}_t, E_t), \quad \Phi = \phi_{10} \circ \dots \circ \phi_0,$$

where E_t is the events of the tick and Φ is the concatenation of tick-orchestrator phases (13 phases, a pure reducer): phases produce deltas, state mutation is performed by a single applicator (`StateApplicator`), and the commit is atomic and unique per tick.

The key ontological invariant (S188) reads: *a communicative claim is never world truth and never directly mutates the relationship store*. A belief is a state of an agent’s mind, not a fact of the world; a lie accepted on faith changes the listener’s psyche but not the history of the universe. All further mechanisms rest on this axiom.

3.2 ExperienceTrace: experience with first-class provenance

The unit of interpreted perceived experience is `ExperienceTrace` (contract E1.0)—an immutable record with mandatory provenance. The `source_type` field takes three values: `PERCEPTION` (the agent perceived it directly), `TESTIMONY` (learned from another agent—then `source_id` points to the narrator), and `INTROSPECTION` (one’s own action or state). The triad of participants—`actor_id` (who committed the event), `owner_id` (whose experience it is), and `source_id` (how the owner

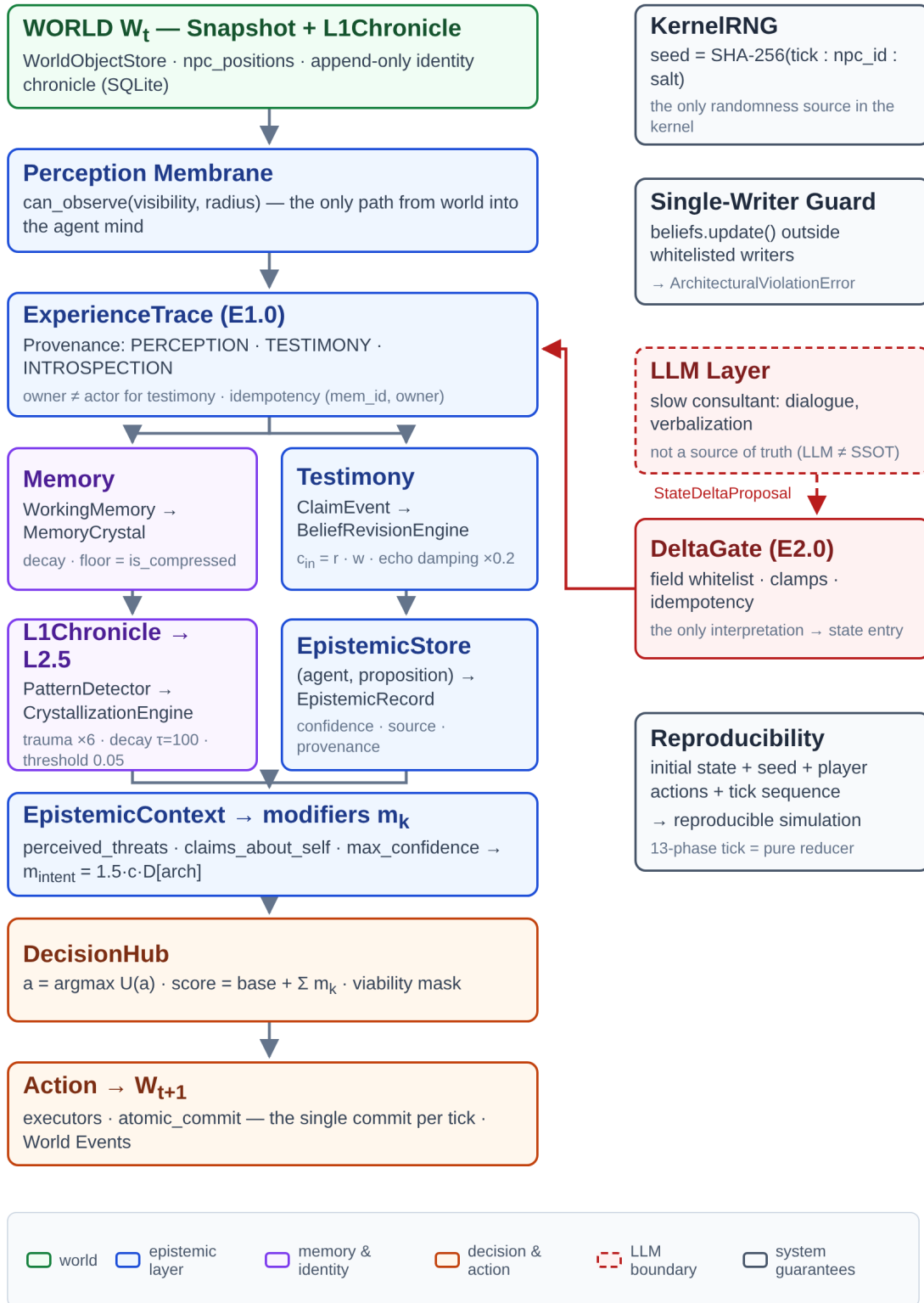


Figure 1: The ENIGMA architecture: the causal pipeline (left) from the world W_t through the perception membrane and ExperienceTrace to beliefs, decisions, and actions; the LLM boundary with the DeltaGate gateway (top right) and the system-wide determinism guarantees (right). The red arrow is the single legal “interpretation → state” entry.

knows)—makes *owner* $\neq$ *actor* a principal case: the experience of a witness to someone else’s theft and the experience of the thief are different records with different trust semantics.

Idempotency is guaranteed by record identity: (episode, owner), and for testimony (episode, owner, source). Redelivery of the same event does not create duplicate experience. Signal fields (valence, arousal, novelty, relevances) are projections filled by a resolver rather than by the record itself: ExperienceTrace is not a source of truth (the SSOT remains event memory) but an interpretation over it.

3.3 The perception membrane

The world enters an agent’s “mind” only through the perception membrane. Event visibility is determined by the `can_observe` function: private events are seen only by the source, whispers by the source and the addressee, public events by everyone within the hearing radius (comparison of Euclidean distance against the radius). The communication-event subscriber forms the set of listeners by a spatial query of the scene: the addressee of an utterance remains its semantic recipient, but physically only those nearby hear it; “telepathy” (transferring beliefs without physical contact) is forbidden. Sight is likewise limited by the field of view, and the embodiment of all agents, including the player, obeys uniform physical laws.

3.4 DeltaGate: the single entry into state

Between any interpretation (including an LLM-generated one) and agent state stands the `DeltaGate` gateway (contract E2.0). Its invariant is stated directly in the code: without applying the gateway, no interpretation—including an LLM-generated one—changes a single field of the psyche. The mechanism is threefold:

1. **Field whitelist.** Only two fields are legal: `threat_gradient` with range $[-1, 1]$ (consumer: `perceptual_kernel`) and `danger_belief` with range $[0, 1]$ (consumer: `r7_beliefs`); an attempt to write any other field is rejected with a logged warning.
2. **Clamps.** The value is clamped into the admissible range before dispatch to the consumer.
3. **Idempotency.** The key (trace id, field) is remembered; a repeated delta of the same trace is silently skipped.

Every application publishes an `EXPERIENCE_DELTA_COMMITTED` event with full causal tracing (trace, causal parent, field, value, consumer)—the single output for the chronicler. The role of the LLM in this architecture is that of a “slow consultant”: the language model handles dialogue and verbalization, the fast world (ticks) never waits for slow intelligence (the two-speed doctrine), and no generated utterance becomes a world fact without deterministic authorization of the mutation. The LLM is not the substrate of ENIGMA’s intelligence; the deterministic core owns state, time, causality, memory, beliefs, and decisions.

3.5 Belief revision: source reliability and echo damping

Testimony is processed by `BeliefRevisionEngine`—a deterministic engine that uses no LLM and does not mutate relationships. Source reliability r is computed by a provider (the sole owner of the formula) from the trust τ of the listener toward the speaker:

$$r(\tau) = \begin{cases} 0.9, & \text{direct observation (DOR),} \\ \text{clamp}(\tau/100, 0, 1), & \tau \geq -30, \\ -\frac{|\tau| - 30}{70}, & \tau < -30. \end{cases}$$

The prior reliability of an unknown source ($\tau_0 = 50$) is 0.5; an enemy ($\tau = -100$) yields $r = -1$ (their words act in the opposite direction, but not below zero). Continuity at $\tau = -30$ and monotonicity everywhere are checked by the INV-EPISTEMIC-TRUST-MONOTONICITY invariant.

The incoming confidence $c_{\text{in}} = r \cdot w$ (w is the claim weight, 1.0) updates the record along three branches:

$$c' = \begin{cases} \max(0, c_{\text{in}}), & \text{new record,} \\ \text{clamp}(c + 0.2 c_{\text{in}}), & \text{same source (echo),} \\ \text{clamp}(c + c_{\text{in}}), & \text{independent source.} \end{cases}$$

The $\times 0.2$ damping of repeats from the same narrator prevents confidence inflation by the echo of a single rumor; the full increment is available only to independent confirmation. Each `EpistemicRecord` stores provenance: the source, the id of the originating claim, and the ticks of first observation and last update.

3.6 Belief crystallization: asymmetric trauma and entropy

Long-term interpersonal beliefs are formed by the crystallization engine `BeliefCrystallizationEngine` (layer L2.5) from aggregated pattern statistics (L1.5 `PatternDetector` over the append-only L1 chronicle). The statistics are pure: cumulative effect `cum` and behavioral variance, with no psychological fields; psychology is added only by the crystallizer. The base weight of an episode,

$$w_{\text{base}} = \min(|\text{cum}|/10, 1) \cdot s, \quad s = \begin{cases} s_{\text{fear}}, & \text{cum} < 0, \\ s_{\text{signif}} + s_{\text{desire}}, & \text{cum} > 0, \end{cases}$$

is modulated by personality (sensitivity to threat or to significance). The weight update rules are:

$$\text{confirmation: } w' = \min(w + w_{\text{base}}, 1); \quad \text{refutation: } w' = w - 6 w_{\text{base}},$$

where the trauma multiplier equals 6 (the constant `TRAUMA_MULTIPLIER`, ADR-O-307): refuting trust is six times stronger than confirming it. If the weight of a refuted belief drops to zero, the old one is erased and a new one of the opposite sign may form; a new belief is created when $w_{\text{base}} > 0.05$.

The second component of the dynamics is entropy decay:

$$w(t) = w(t_0) e^{-\Delta t/100}, \quad \tau_{\text{decay}} = 100 \text{ ticks},$$

with a forgetting threshold of $w \leq 0.05$: a belief not reinforced by new episodes loses half its weight in $\tau \ln 2 \approx 69.3$ ticks and is erased after roughly 300 ticks. Crystallized beliefs feed the L3 layer (effective drives) and, through them, decision utility.

3.7 Determinism: KernelRNG, single-writer, and the append-only chronicle

Reproducibility is ensured at three levels. First, the only source of randomness in the core is `KernelRNG`, whose seed is computed deterministically:

$$\text{seed} = \text{SHA-256}(\text{tick} : \text{npc_id} : \text{salt}).$$

Identical pairs $(\text{tick}, \text{npc_id})$ yield identical sequences, while different agents or ticks yield independent ones; direct use of `random.*` and `time.time()` in the core layer is prohibited by a dedicated linter. The target reproducibility formula is: *initial state + seed + player actions + tick sequence* $\rightarrow$ *reproducible simulation*.

Second, writes to critical state are protected by single-writer guards: the `update()` method of the belief state checks the calling module against a whitelist of legal writers; a direct write bypassing `BeliefTransitionEngine` raises `ArchitecturalViolationError`—an attempt to circumvent the guard becomes a failing test rather than silent state corruption. Third, the identity chronicle `L1Chronicle` is an append-only SQLite journal with a unique constraint on the quintuple (campaign, target, tick, source, event type): history only grows; archiving old events moves records into an archive table, deleting nothing.

The overall flow diagram is shown in Fig. 1: a vertical causal pipeline from the world through the membrane and experience to beliefs, decisions, and actions; on the right, the LLM boundary with the `DeltaGate` gateway and the system-wide determinism guarantees.

Table 1: Executable epistemic contracts of SUPERBOX (snapshot V.0.5.4.0.6, fresh-extraction protocol).

Contract	Property under test	Verdict
Determinism core (IPT)	45 core invariant probes: commit cardinality, state round-trip, FSM zombies, object topology	green
Epistemic core (S-002)	revision is deterministic, provenance preserved, truth and belief separated, independent confirmation	green
Single-writer (D3)	a direct belief write bypassing the engine raises <code>ArchitecturalViolationError</code>	green
Observation channel (O-360)	LOS, distance, and walls: observation $c=0.9$; no telepathy; world truth unchanged	green
Dispositions (S211)	one belief $\rightarrow$ different modifier dictionaries per archetype; deterministic	green
Causality: DecisionHub	modifiers ($c=0.9$, guard) flip the intent warn $\rightarrow$ report, score 0.225 $\rightarrow$ 2.093	green
Causality: full loop (CAUSAL-POS)	THEFT $\rightarrow$ 4 observers ($c=0.9$) $\rightarrow$ disposition shifts $\rightarrow$ intent change	green
Causality: testimony (S-004/010)	a rumor from an agent with trust -0.4 yields $c=0$ and no decision change	red ^a
Membrane: communication (S-008)	an addressee outside the hearing radius receives no belief	red ^b
Persistence (S-009)	beliefs survive save/load and loop restart	green
Player (S-014/015)	NPC lines $\rightarrow$ player beliefs (friend 0.8, enemy 0.0)	green
ACCUSE gate (S211)	a belief ≥ 0.5 is a gameplay capability; a false accusation passes	green
Second order (S-013)	an agent believes that another agent asserts P	needs LLM ^c

^a the chain works, but the causal force of testimony is gated by source reliability (see the positive control); ^b the default event radius 999 admits a distant listener; the observation membrane passes all checks; ^c the event-to-state wire requires a live LLM server.

4 Executable Epistemic Contracts

A claim about the ENIGMA system counts as implemented only when an executable scenario passes: *claim* $\rightarrow$ *code* $\rightarrow$ *call path* $\rightarrow$ *test* $\rightarrow$ *observable result*. The SUPERBOX scenario repository simultaneously serves as the acceptance gate for epistemic claims: each scenario proves one causal contract of the chain “belief $\rightarrow$ decision $\rightarrow$ action $\rightarrow$ world event” and terminates with a binary verdict. What would count as project failure: all agents see objective truth; beliefs are aliases of world state; dialogue asserts a belief the engine does not use; the LLM invents a fact with no path into state.

The reference “Double Truth” scenario (SUPERBOX-001/003) is structured as follows. World: three agents A , B , C ; A and B are not on good terms. Event: A steals an item from B ; B witnesses the theft; C is not present. Communication: A tells C that B stole it. Expected outcome: B believes (from perception) that the culprit is A ; C believes (from testimony, trusting A) that the culprit is B ; in one and the same world, B and C act differently— C turns hostile toward B because of a false belief. Crucially, the scenario never injects the belief directly: only the cause is injected—the communication event.

The key contracts are summarized in Table 1. The “verdict” column reflects the results of our runs of snapshot V.0.5.4.0.6 under a fresh-extraction protocol (Section 5); historical green verdicts from the authors’ environment are documented in the project’s mutation log (S186–S211).

The methodological difference between SUPERBOX and ordinary unit tests is the level of the claim. A unit test checks a function; a SUPERBOX contract checks an *engineering statement about the architecture*: “belief divergence is causally efficacious” cannot be verified by a function, but it can be verified by running a world. This is why the verdicts of Table 1 read as an epistemic audit of the system, and the red cells as equally valid results: they indicate exactly *where* the causal chain is broken in a given snapshot, as demonstrated in Section 5.9.

5 Experimental Evaluation

5.1 Experimental setup

All results were obtained by running snapshot V.0.5.4.0.6 in an isolated Linux environment without an LLM server. This condition is essential: the deterministic core is designed to survive the unavailability of the “slow consultant”—the tick continues, and LLM-dependent subtasks degrade with an observable logged error (no silent failures). The scenarios themselves run through the standard entry points of the project. A second condition is the *fresh-extraction protocol*: every family of runs starts from a pristine unpacking of the release archive. The sandbox persists world state between runs (tick counter, positions, relationships), and accumulated history flips scenario verdicts—an effect documented in Section 5.9; fresh extraction eliminates it, and every number below is reproducible from that state.

5.2 Deterministic core

The invariant probe suite (IPT) passed in full: **45 of 45** probes in approximately five seconds. The probes include property-based tests (NPC state round-trip on 200 examples, degenerate-coordinate rejection on 100 examples, FSM-zombie detection on 100 examples), commit cardinality (exactly one commit per tick), world object topology (24 live objects), and behavioral locks. A `KernelRNG` microcheck: sequences for the identical pair (42, `npc_a`) matched bit for bit (0.630546, 0.838969, ...), and were independent for the pair (42, `npc_b`) and the next tick (43, `npc_a`). The single-writer guard was verified by failure: a direct write to `state.beliefs` from an arbitrary module raises `Architectural ViolationError` (test passed). The invariant counter is monotone across the project’s development history—39/39 at S190, 44/44 at S204, 45/45 in the studied snapshot, and 45/45 in the authors’ environment log.

5.3 Epistemic core

Scenario SUPERBOX-002 passed all four checks. (1) A belief was created from the `ClaimEvent` with exact arithmetic: confidence = $r \cdot w = 0.8 \cdot 1.0 = 0.80$; provenance (source, claim id) preserved. (2) The iron invariant: after accepting a lie on faith, the objective world truth remained unchanged—truth and belief remain separated states. (3) Determinism: re-revising the same claim produced a record bit-identical to the first. (4) Independent confirmation by a second source ($r = 0.4$) raised confidence $0.80 \rightarrow 1.00$ (clamped at 1.0), whereas an echo of the same narrator would have added only 0.08. The integration of the core into the causal machine is confirmed by separate scenarios: a belief is created by publishing `COMMUNICATION_CLAIM` on the `EventBus` (provenance preserved, world truth untouched); the projection `EpistemicStore` $\rightarrow$ `EpistemicContext` (threats, allies, violations) is deterministic and does not mutate the store; and the composition of `EpistemicContext` into `DecisionContext` preserves encapsulation—`DecisionHub` reads threats without knowing that Proposition or the store exist.

5.4 Observation channel and dispositions

The second channel through which beliefs arise is direct observation of world events (ADR-O-360; a five-check scenario, all green). A theft event produces the belief (`player`, `STOLE`, `target`) with direct-observation reliability $r = 0.9$ in a witness within line of sight; a witness at distance $13.0 > 10.0$ receives nothing (telepathy forbidden), as does a witness behind a wall (distance 6.0, line of sight blocked); world truth is not mutated in the process. Repeated observation of the same event by the same witness strengthens the belief $0.9 \rightarrow 1.0$ (same-source boost), and hostile testimony layered over observation weakens it $1.0 \rightarrow 0.7143$ —exactly by the ADR-O-357 formula, verified by predicting the value before the run.

The disposition layer (S211, six checks) closes the projection from context to decision modifiers. One and the same `EpistemicContext` (threat, $c = 0.9$) yields four *different* modifier dictionaries by archetype: the guard gets `report` = 1.89 > `warn` = 0.675; the maid, `spread_rumor` = 1.755—the

maximum of her profile; the merchant, `warn` = 1.89; and the thief receives no `report` modifier at all (the criminal’s silence). A call without an archetype reproduces the S198 legacy dictionary byte for byte, and a repeated call is identical (determinism). One belief—different deeds by nature: the monoculture of reactions is dead architecturally, not by prohibition.

5.5 Belief-to-decision causality: three levels of confirmation

The central architectural claim—the causal efficacy of belief—is confirmed at three levels of isolation. *Level 1: direct DecisionHub call.* Agent `guard_borko`, with an identical seed and drives, decides `warn` with score 0.225 without epistemic modifiers; with modifiers computed from a context (threat, $c = 0.9$, guard archetype: `report` = 1.89), it selects `report` with score 2.093. The decision changed by a single input: the belief.

Level 2: the full live loop (positive control). A single world theft event (actor: the thief) is injected between ticks in the tavern. The LOS membrane determines the observers: four agents (blacksmith, maid, merchant, player) form the belief (thief, `STOLE`, apple) with $c = 0.90$; the guard, separated by a wall, does not (distance $8.9 < 10$, but the line of sight is blocked); the thief himself is excluded as the actor. On the next tick the witnesses’ decision scores shift by their disposition modifiers: the blacksmith (commoner) `warn` $0.004 \rightarrow 1.452$ (modifier 1.35); the merchant `warn` $-0.098 \rightarrow 1.89$ (modifier 1.89); the maid `talk` $0.181 \rightarrow 0.998$ (modifier 0.81)—and her intent changes `idle` $\rightarrow$ `talk`. The actor’s scores remain bit-identical; non-witnesses show only a small non-epistemic shift (Section 5.9). Decision divergence is produced by belief divergence in one and the same world.

Level 3: the testimony channel and its precondition. A rumor (`COMMUNICATION_CLAIM` from the thief) reaches six listeners, but belief is a function of trust in the narrator (Section 5.7): the guard, who suspects the thief (trust -0.4), obtains $c = 0.00$, and the decision does not change (S-004/S-010—a red verdict). This is not a broken belief-to-decision wire—it is confirmed by three independent runs—but the operation of its precondition: the causal force of testimony is gated by source reliability. Levels 1–2 prove the wire; level 3 measures the gate.

5.6 Micro-dynamics of beliefs

Table 2 summarizes measurements of the isolated subsystems (all numbers are outputs of runs, not formulas from documentation). The trauma asymmetry manifests in two ways: the pure engine multiplier is 6, while the effective ratio of shifts at branch sensitivities 0.25/0.50 is 3—a refutation with $|\text{cum}| = 2$ shifts the weight by 0.300, a confirmation of the same strength by 0.100. A trust belief of 0.60 is destroyed by three consecutive negative episodes. Decay: half-life 69.3 ticks, forgetting horizon 299.6 ticks (for $w_0 = 1.0$). Revision: an echo of the same source raises confidence $0.40 \rightarrow 0.48$; an independent source, $0.40 \rightarrow 0.80$; a hostile source ($r = -1$) yields zero confidence.

5.7 A live rumor cascade

A run of the live scene (tavern, 6 NPCs + player) with the injection of a single communicative claim “the merchant stole the apple” from a lying agent produced exactly one record per listener in the epistemic store—and *different* confidence for different listeners (Table 3): one rumor, one source, six epistemic outcomes. Each listener’s confidence is exactly their source reliability $r(\tau)$: default trust 0.5 for agents with no relationship history with the liar; 0.1 for the maid (his handler, but distrusting); 0 for the innkeeper ($\tau = -0.2$) and the guard ($\tau = -0.4$, who suspects the liar). The subject of the slander, with default trust, half-believes the accusation about himself (0.5)—the self-referential channel does not override trust. This is a computational implementation of what social epistemology calls the distribution of belief over a trust network—and all of it is a deterministic function of the relationship state, reproducible tick after tick.

Table 2: Micro-dynamics of beliefs: measurements of isolated subsystems (V.0.5.4.0.6, no-LLM environment).

Measurement	Result
Confirmation (trust, $w=0.60$, $ \text{cum} =2$)	$0.594 + 0.100 = 0.694$
Refutation (same strength)	$0.594 - 0.300 = 0.294$
Pure engine asymmetry	$6.0\times$; effective $3.0\times$
Destruction of trust = 0.60 ($ \text{cum} =2$)	3 negative episodes
Weight half-life	$\tau \ln 2 = 69.3$ ticks
Forgetting horizon ($w_0=1.0 \rightarrow 0.05$)	$\tau \ln 20 = 299.6$ ticks
Echo of the same source ($c=0.40$, $r=0.40$)	$0.40 \rightarrow 0.48$
Independent source ($r=0.40$)	$0.40 \rightarrow 0.80$
Hostile source ($r = -1.0$)	$c = 0.00$
KernelRNG (42, npc_a), two runs	sequences identical

Table 3: The cascade of a single rumor “the merchant stole the apple” (source: agent *A*, a liar): listener confidence as a function of trust in the source.

Listener	Role in the scene	Confidence
Merchant Goran	subject of the slander (about self)	0.50
Blacksmith Orm	witness nearby	0.50
Player	observer	0.50
Maid Lucy	witness at the radius edge	0.10
Innkeeper Tornin	distrustful listener ($\tau=-0.2$)	0.00
Guard Borko	suspects the liar ($\tau=-0.4$)	0.00

5.8 Persistence and player-facing closure

Persistence (S-009, green): lie injection, two ticks, scene save, **GameLoop** restart, load—the belief is restored with the same confidence (0.50), and the epistemic state before and after the restart is identical. Bit-level parity also holds for the full store: 8 in-memory records project into 8 records of the serialized snapshot with the same confidences—the “state $\rightarrow$ snapshot” boundary does not lose epistemics.

Player-facing closure (S-014/S-015, green): NPC lines produce player beliefs as a function of trust—a friend ($\tau = +50$) yields $c = 0.80$, an enemy ($\tau = -50$) yields $c = 0.00$; the runtime pipe **DialogueMaterializer** $\rightarrow$ **EventBus** $\rightarrow$ **EpistemicStore**[player] works without manual calls ($c = 0.96$), and the full causal pipe **TaskScheduler** $\rightarrow$ **executor** $\rightarrow$ **materializer** $\rightarrow$ **EventBus** $\rightarrow$ listeners’ beliefs produces three records in one tick: the player (0.99), the target witness (0.50), and the speaker himself (0.50).

The accusation gate (S211, five checks, green) takes the principle into gameplay: the **ACCUSE** action is available to the player only with a belief $c \geq 0.5$ about the accused subject. A weak rumor ($c = 0.4$) is blocked, as is the absence of belief, but a *false* accusation of an innocent under firm belief (0.9) passes: the engine is not an arbiter of truth—social consequences (**fear** +25, **trust** −15) are applied to the target, and the divergence of belief from truth is resolved by the agents themselves. Belief is a capability in the literal sense: the player acts from knowledge, including mistaken knowledge.

5.9 An honest audit: red verdicts as diagnostics

Six groups of checks returned red verdicts in the studied snapshot. We treat these failures as first-class results. In a system that claims executable epistemic contracts, a red verdict is more valuable than a silent green: it indicates exactly where the causal chain is broken under the given fixture (Table 4).

Table 4: Red verdicts of snapshot V.0.5.4.0.6 (fresh-extraction protocol) and their diagnosis.

Scenario	Diagnosis
S-004/S-010 (testimony)	the guard’s trust in the liar -0.4 yields $c = 0$; empty context, empty modifiers, no decision change—the chain is confirmed by the positive control and gated by trust
S-007/S-008 (membrane)	the default event radius 999 admits a distant listener ($c = 0.10$ at distance 12.6); the observation membrane passes 5/5
S-009-gap (DecisionHub)	a fixture without <code>max_confidence</code> $\rightarrow$ zero boost $\rightarrow$ identical decisions; with a full context, $0.225 \rightarrow 2.093$
S-005/S-006 (attribution)	<code>base + mod $\neq$ final</code> : an unattributed -0.024 deformation from non-epistemic channels
E2.0-c and the LLM family	“the event-to-state wire is dead” without the LLM environment—the degradation gateway does not compensate for the missing interpretation
Run state	accumulated world state flips verdicts across runs (a “WARN without target” crash); the fresh-extraction protocol is mandatory

Causality via testimony (S-004/S-010). The belief formed with confidence 0.00: the guard’s trust in the liar is negative in the world fixture ($\tau = -0.4$, “suspects him of crimes”), the revision honestly nullified the rumor, the context is empty, there are no modifiers, and the decision did not change. This is not a break in the belief-to-decision chain—it is confirmed by three independent runs (Section 5.5: the direct `DecisionHub` call, the observation positive control, and the S211 dispositions)—but the operation of its precondition: the causal force of testimony is gated by source reliability. The scenario assumes a fixture with positive trust.

The communication membrane (S-007/S-008). A manually injected event receives the default radius of 999: everyone “hears” it, including the addressee at distance 12.6 (belief $c = 0.10$). Meanwhile the observation-channel membrane (ADR-O-360) passes all five checks: distance, walls, and self-exclusion work; there is no telepathy. The verdict split by channel: the physics of observation is proven; the communication event radius is not isolated from world state.

The DecisionHub gap (S-009-gap). The test constructs an `EpistemicContext` without `max_confidence` (default 0.0); the modifier boost is $0 \cdot 1.5 = 0$, and both calls return identical decisions. We reproduced the red verdict and refuted its diagnosis: with a correctly filled context, `DecisionHub` applies the modifiers and flips the intent (Section 5.5, level 1). The red verdict recorded a defect in the test’s own fixture—which is also useful contract work: the oracle failed exactly where the context projection is incomplete.

Modifier attribution (S-005/S-006). The bit-level check `base + mod = final` does not hold: a small unattributed deformation (-0.024 on every intent) arrives from non-epistemic channels of the event (the affective phase, the threat causal slice). The term is deterministic but not attributed by contract—the boundary of modifier responsibility is blurred.

LLM-dependent scenarios (S-011/S-012/S-013, GORAN, production). Without a live LLM server the “event $\rightarrow$ state” wire does not close: degradation is reported to the log (no silent failures), but `NPC_SPOKE` events are not produced, and the second-order observation and the vertical slice do not assemble.

Run-state accumulation. The sandbox persists world state between runs: after roughly a dozen accumulated ticks, a tick crashes with “intent WARN without target”—the accumulated world drives an agent into a state the scenario fixture did not anticipate. All verdicts in this paper were obtained under the fresh-extraction protocol; without it, verdicts reproduce poorly. This is a direct demonstration of the “oracle depends on world state” thesis—at the level of the instrumentation itself.

We report both series—the reds obtained in this snapshot and the greens recorded in earlier author environments (the mutation log: the membrane S192.1, decision divergence S194, action causation S195, world event causation S196, second-order observation S198a, the observation channel S207 5/5, the dispositions S211 6/6, GORAN 12/12)—because their juxtaposition is the point of the method. A system

that only ever reports green is unverifiable. Executable contracts are useful precisely when they can fail loudly and point to a concrete break.

6 Discussion and Limitations

What engineering-verifiable epistemics buys. The main practical result of the architecture is the ability to answer epistemic questions by inspecting the run rather than by re-querying the model. Where an agent learned a fact (the trace provenance), how confident it is (record confidence with source provenance), why a decision changed (modifiers computed from beliefs, with the triggering proposition traced)—all of these are readable, deterministic functions of state. For researchers, this turns “behavioral” claims about agents into measurable ones; for game developers, it offers a causal alternative to the “the NPC just knows” script. It also matters that determinism does not make the world predictable for the player: mechanical reproducibility and informational unpredictability are orthogonal properties separated by the observer’s epistemic boundary.

Asymmetry as an engineering decision. The trauma multiplier of 6 is not derived from psychological data; it is a calibration constant formalizing negativity bias in computable form. Its value is checkable: the ratio of refutation and confirmation shifts is measured directly (Section 5), and the branch sensitivities modulate the effective asymmetry ($3\times$ at $s_{\text{fear}}=0.25$, $s_{\text{signif}}+s_{\text{desire}}=0.50$). Such explicitness favorably distinguishes the system from LLM agents, where “grudge retention” is an implicit property of prompts.

Limitations. We list them explicitly, following the principle “do not claim more than the scenarios prove”. (1) A full Theory of Mind is not claimed: second-order observation is documented in the authors’ environment (S198a) but requires a live LLM server in the studied snapshot and does not reproduce without one; the ASSERTS predicate exists, deeper cascades do not. (2) Several scenario oracles still depend on world state (relationships, positions, default radii, ticks accumulated across runs). As the audit of S-004 and S-008 shows, this makes the verdict sensitive to the fixture. Isolating contracts behind explicit world preconditions remains future work. (3) Two legal writers coexist in the belief state (the reactive branch and the coherence aggregator)—the documented R8 compromise without merge semantics. (4) Async-interleaving nondeterminism (DEBT-QUIESCE) remains an open reproducibility debt. (5) Scale: one location, 6 NPCs; scaling to hundreds of agents was not measured. (6) Constants are calibrated heuristically; no quantitative comparison with baseline approaches (LLM agents with memory) was conducted—the work is positioned as a systems contribution, and its evaluation is the feasibility and reproducibility of the contracts, not behavioral superiority.

7 Conclusion

We have presented a deterministic multi-agent architecture in which the divergence of agent beliefs is a first-class, causally efficacious, and executably verifiable property of the simulation rather than a side effect of language generation. The architectural foundation consists of experience with first-class provenance, the perception membrane, the DeltaGate gateway isolating the LLM from the causal core, and asymmetric belief crystallization; the methodological foundation consists of executable epistemic contracts that turn architectural claims into binary reproducible verdicts. The runs confirmed a working core (45/45 invariants), the determinism and measurability of epistemic micro-dynamics, the distribution of a single rumor’s confidence over a trust network, the belief-to-decision causality at three levels of isolation, and—equally—the informativeness of red verdicts as an audit instrument. Future work: behavioral closure, unified affective appraisal, isolation of scenario oracles from world state, and quantitative calibration of the constants. We regard ENIGMA not as another LLM sandbox, but as an attempt to make agent epistemics engineer-verifiable. The invitation to the community is concrete: the instrument that produced the green and red verdicts in this paper is public. Check the claims with the contracts.

Availability

The complete source code, the mutation log, architectural ADRs, and the SUPERBOX scenarios are available in the project repository: <https://github.com/4Um4/Enigma/> (branch V.0.5.4.0.6_Kushaem_1). All runs in this paper execute in a no-LLM environment from a freshly unpacked release archive. A quick core check (about five seconds, no LLM required):

```
python backend/tests/IPT.py
```

The contract scenarios of Sections 4 and 5 (from the `backend` directory, each from a fresh extraction):

```
python tests/sandbox/SUPERBOX/scenarios/epistemic_core_test.py
python tests/sandbox/SUPERBOX/scenarios/epistemic_observation_test.py
python tests/sandbox/SUPERBOX/scenarios/epistemic_dispositions_test.py
python tests/sandbox/SUPERBOX/scenarios/epistemic_persistence_test.py
python tests/sandbox/SUPERBOX/scenarios/epistemic_player_belief_test.py
python tests/sandbox/SUPERBOX/scenarios/accusation_gate_test.py
python tests/sandbox/SUPERBOX/scenarios/epistemic_causal_test.py
```

The positive causality control (Section 5.5, level 2) is published alongside the paper: an injection of a THEFT event through the standard `EventBus` between ticks, followed by a comparison of the control/treatment branch intents. The micro-measurements of Section 5 are reproduced by isolated runs of the engine modules: belief crystallization, belief revision, and `KernelRNG`.

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